

CHARLES JOSEPH

PhD Scholar | Climate Science & Modelling

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Academic researcher and university educator with expertise in machine learning, deep learning, spatiotemporal modelling, and decision-support systems. Experienced in undergraduate teaching, curriculum development, interdisciplinary research, and publishing in high-impact journals. Passionate about translating artificial intelligence research into real-world solutions through industry collaboration and student-centred education.

Education

- 2024 - present **PhD, Information Systems, University of Southern Queensland, Australia.**
Thesis: *Explainable Deep Learning for Climate Risk Modelling and Basis Risk Reduction in Innovative Financial Tools for Agribusiness*
Application domains: Climate risk, agribusiness, financial innovation
- 2009 - 2013 : **BSc, Computing and Information Systems, Sabaragamuwa University of Sri Lanka, Sri Lanka.**
Thesis: *Machine Learning-Based Emotion Classification of Musical Audio Signals*
CGPA: 3.60/4.00

Work Experience

Queensland University of Technology, Australia (QUT)

- Mar - May 2026 **Research Assistant.**
- Collaborated with interdisciplinary teams to deliver reports, publications, and presentations in the field of Computational Mechanics.

University of Southern Queensland, Australia (USQ)

- Nov 2025 – Present **Research Assistant.**
- Develop AI-based models for analyzing large-scale environmental and temporal datasets.
 - Support interdisciplinary research projects involving data processing, evaluation, and reporting.
 - Collaborate with supervisors and researchers to translate analytical results into research outcomes.

Eyeo GmbH, Germany

- Apr - Sep 2024 **Acceptable Ads Monitoring Specialist.**
- Reviewed websites for compliance with agreed standards.
 - Verified Acceptable Ads delivery across test environments and platforms.
 - Developed test cases in collaboration with TAM/AM/AA Monitoring teams.
 - Conducted pre-release, circumvention, and legal compliance checks.
 - Monitored and tested DNS-based ad filtering solutions to prevent service disruptions.

Sabaragamuwa University of Sri Lanka, Sri Lanka

- Oct 2021 - May 2023 **Lecturer in Computer Science, Department of Physical Sciences and Technology.**
- Designed and delivered undergraduate Computer Science and Information Technology courses using student-centred teaching approaches.
 - Developed curriculum, lesson plans, practical laboratory activities, and assessment tasks aligned with learning outcomes.
 - Supervised undergraduate student projects and provided academic mentoring.
 - Assessed student performance and provided constructive feedback to support learning.
 - Collaborated with academic staff to continuously improve curriculum quality and student engagement.
- Oct 2020 - Sep 2023 **Visiting Lecturer in Computer Science, Center for Open and Distance Learning.**

- Oct 2020 - Sep 2021 **Temporary Lecturer in Computer Science**, Department of Physical Sciences and Technology.
- Apr 2020 - May 2023 **System Administrator**, Centre of Computer Studies.
- Led the implementation of a new Learning Management System (LMS).
 - Delivered training sessions for faculty, staff, and students.
 - Provided technical support and troubleshooting.
 - Managed digital learning resources and maintained course materials in collaboration with academic staff.
 - Analyzed LMS usage data and prepared reports.
- Jan 2020 - Sep 2023 **Temporary Demonstrator in Computer Science**, Department of Computing and Information Systems.
- Eyepax IT Consulting, Sri Lanka
- Nov 2017 - Apr 2018 **Trainee Software Engineer**.
- Designed and developed PHP-based applications.
 - Developed and executed unit test cases to ensure software reliability and functionality.
 - Implemented API integrations to enhance system connectivity and performance.
 - Worked with JavaScript frameworks, including Angular and Node.js.

Publications

Journal Articles

- 2026 **Charles Joseph**, Qingxia Wang, Mushtaq Shahbaz, Li Yan, Jonathan Barratt, and Tim Barratt. Learning drought memory for wheat yield anomaly prediction: A residual attention LSTM using multi-scale drought indicators. *Technological Forecasting and Social Change (Q1, Impact Factor: 13.5)*. Elsevier, 2026. (Submitted).
- 2026 **Charles Joseph**, Qingxia Wang, Mushtaq Shahbaz, Li Yan, Jonathan Barratt, and Tim Barratt. Deep learning the droughts: Spatiotemporal monitoring of protracted dry conditions in queensland, australia. *Science of the Total Environment (Q1, Impact Factor: 8.0)*. Elsevier, 2026. <https://doi.org/10.1016/j.scitotenv.2026.181564>.
- 2026 **Charles Joseph**, Qingxia Wang, Mushtaq Shahbaz, Li Yan, Jonathan Barratt, and Tim Barratt. Advancing weather index insurance for climate risk management: A review of modelling techniques and implementation strategies. *Environmental Impact Assessment Review (Q1, Impact Factor: 12.2)*, volume 119, page 108376. Elsevier, 2026. <https://doi.org/10.1016/j.eiar.2026.108376>.
- 2024 Thuseethan Selvarajah, Vigneshwaran Palanisamy, **Joseph Charles**, and Wimalasooriya Chathrie. Siamese network-based lightweight framework for tomato leaf disease recognition. *Computers (Q1, Impact Factor: 5.2)*, volume 13, page 323. MDPI, 2024. <https://doi.org/10.3390/computers13120323>.
- 2024 Hewarathna Ashen Iranga, Hamlin Luke, **Joseph Charles**, Vigneshwaran Palanisamy, George Romiyal, Thuseethan Selvarajah, Wimalasooriya Chathrie, and Shanmugam Bharanidharan. Change detection for forest ecosystems using remote sensing images with siamese attention U-Net. *Technologies (Q1, Impact Factor: 5.2)*, volume 12, page 160. MDPI, 2024. <https://doi.org/10.3390/technologies12090160>.

In Conference Proceedings

- 2021 **Joseph, Charles** and Lekamge Sugeeswari. An ensemble learning approach for automatic emotion classification of sri lankan folk music. In *Proceedings of International Conference on Sustainable Expert Systems: ICSES 2020*, pages 289–302. **Springer Nature**, 2021. https://doi.org/10.1007/978-981-33-4355-9_23.
- 2020 **Joseph, Charles** and Lekamge Sugeeswari. Machine learning for emotion classification of sri lankan folk music. In *14th Conference on Industrial and Information Systems (ICIIS)*, pages 360–365. **IEEE**, 2020. <https://doi.org/10.1109/ICIIS47346.2019.9063313>.

Miscellaneous Experience

Awards and Competitions

- 2025 Participant, **Adaptive Hackathon**, organised by Study Queensland and the Department of Primary Industries (DPI), Australia.
- 2025 **First Place Winner**, Innovation Challenges Pitch Competition, Study Queensland, Australia.
- 2024 Recipient of a **Full Postgraduate Scholarship** awarded by the Australian Research Council (ARC) and the University of Southern Queensland, Australia.

Professional Memberships

Associate Member - 4428084, Australian Computer Society (ACS)

Student Member - 22651 Geospatial Council of Australia.

Reviewer

- 2026 **Displays (Q2, IF:3.9)**, Elsevier
- 2025 **Information Processing in Agriculture (Q1, IF:7.4)**, Elsevier
- 2024 **Cogent Engineering (Q2, IF:2.5)**, Taylor and Francis
- 2024 **International Conference on Advanced Research in Computing (ICARC)**, Faculty of Computing, Sabaragamuwa University of Sri Lanka, Belihuloya, Sri Lanka

Position of Responsibility

- 2025 **Technical Support Volunteer**, Geospatial Conference (Locate25), Brisbane, Australia.
- 2025 **Technical Support Volunteer**, SOMETHINGFEST 2025, Brisbane, Australia.
- 2020–2022 **Secretary**, Criterion 01: Programme Design and Development – Task Force for Self-Evaluation Report (SER) compilation for the Program Review (PR), Faculty of Applied Sciences, Sabaragamuwa University of Sri Lanka.
- 2021 **Member**, Publicity and Web Management Committee, 8th International Conference of Sabaragamuwa University of Sri Lanka.

Academic and Professional Experience

- 2026 **Participant, Quantum Computing Summer School**, Co-organised by RMIT School of Computing Technologies and CSIRO Data61 Quantum Systems Team, Australia. Focused on quantum computing software and algorithm development for applications in mathematics, science, and next-generation AI systems.
- 2024 **Visiting Researcher in Rehabilitation and Assistive Robotics**, Assistive Intelligent Robotics Lab (AIROB), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany. Full-time research appointment focusing on rehabilitation robotics and assistive intelligent systems.

Technical Skills

Programming Python (PyTorch, TensorFlow), R, MATLAB, SQL

Research Expertise Artificial Intelligence, Machine Learning, Deep Learning, Spatiotemporal Modelling, Climate Analytics, Data Science, Explainable AI

Teaching Moodle, Canvas, Learning Management Systems

Referees

Dr. Jenny Wang

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Prof. Shahbaz Mushtaq

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